derived keys as outputs. You can derive an almost unlimited number of strong keys from a single high-entropy master key.

> **DEFINITION** A ***key derivation function (KDF)*** allows new keys to be derived from a ***master key*** and a ***context*** string. This allows keys for different purposes to be derived from a single key.

If you recall from chapter 9, macaroons work by treating the HMAC tag of an existing token as a key when adding a new caveat. This works because HMAC is a secure *pseudorandom function*, which means that its outputs appear completely random if you don't know the key. This is exactly what we need to build a KDF, and in fact HMAC is used as the basis for a widely used KDF called *HKDF* (HMAC-based KDF, https://tools.ietf.org/html/rfc5869). HKDF consists of two related functions:

- *HKDF-Extract* takes as input a high-entropy input that may not be suitable for direct use as a cryptographic key and returns a HKDF master key. This function is useful in some cryptographic protocols but can be skipped if you already have a valid HMAC key. You won't use HKDF-Extract in this book.
- *HKDF-Expand* takes the master key and a context and produces an output key of any requested size.

> **DEFINITION** ***HKDF*** is a HMAC-based KDF based on an ***extract-and-expand*** method. The expand function can be used on its own to generate keys from a master HMAC key.